Couette flow

Wiki *Enhanced dissipation for passive scalars*

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The Couette flow is the linear shear $u = (y, 0)$, considered here in the simplest possible domain for this problem, $(x, y) \in \mathbb{T} \times \mathbb{R}$. The equation under study then reduces to

$$\partial_t f + y\partial_x f = \nu \Delta f, \qquad f|_{t=0} = f^{\mathrm{in}}, \qquad (x, y) \in \mathbb{T} \times \mathbb{R}.$$

This is a foundational model in the field, because the same equation governs the vorticity of the two-dimensional Navier–Stokes equations linearized around the Couette flow. In that context Kelvin already solved it explicitly in 1887 (see Kelvin (1887, eq. (34)–(35))). Moreover, the same model, without the diffusion in x, was derived by Kolmogorov (1934) when developing his theory of Brownian motion, and the equation is also of Fokker–Planck type. See also History of the problem.

We use this model to illustrate what typical mixing and enhanced dissipation results look like, and how the two are related. The model is simple enough that each proof technique can be carried out in full. They are all presented below on this single example.

Why this domain? The domain $\mathbb{T} \times \mathbb{R}$ is mathematically convenient: there are no boundaries, and one can take the Fourier transform in both variables.
There is also a more geometric motivation. Consider a circular flow in the plane, $u = rV(r)e_\theta$, for which the equation reads $\partial_t f + V(r)\partial_\theta f = \nu \Delta f$. The log-polar coordinates $X = e^s \cos(\theta)$, $Y = e^s \sin(\theta)$ identify $\mathbb{R}^2 \setminus \{0\}$ with $(\theta, s) \in \mathbb{T} \times \mathbb{R}$, as shown in the figure below.

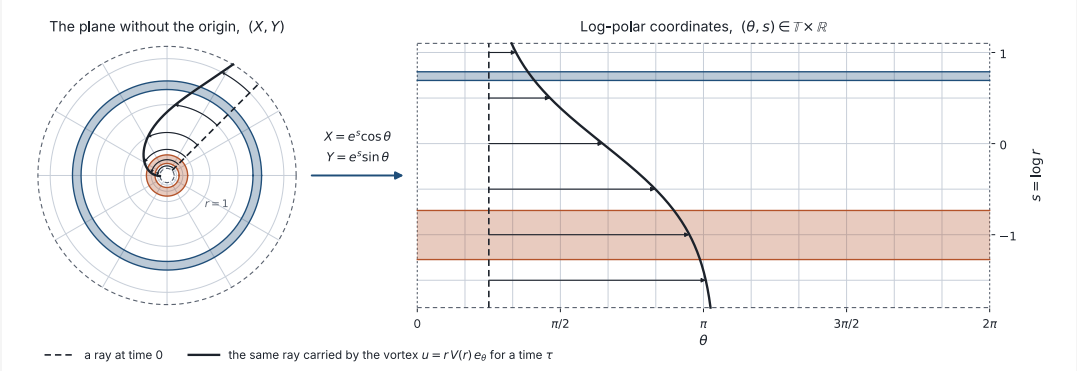


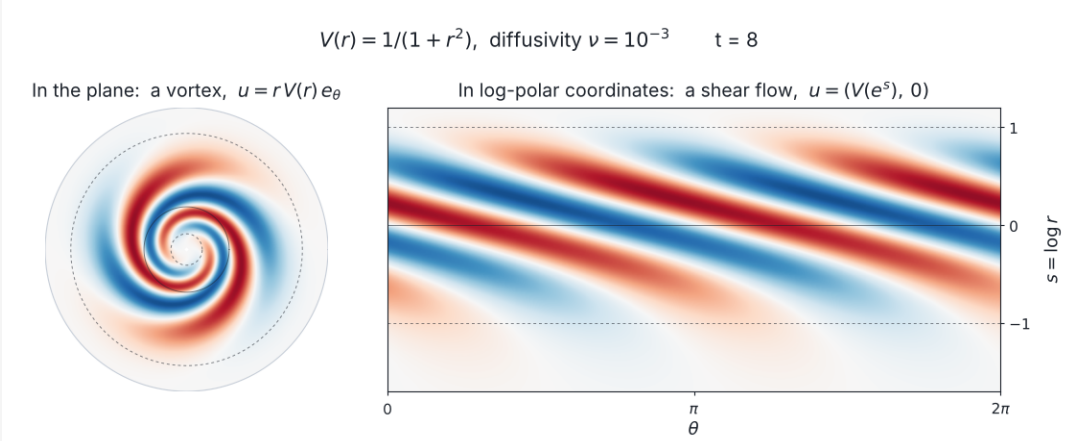
Figure. Left: the plane, with two annuli of the same width and a ray (dashed) carried by a circular flow for some time (solid). Right: the same picture in log-polar coordinates. The annuli become strips, and the circular flow becomes a shear flow.

Note that this is a conformal change of variables, so the Laplacian only gains a scalar

factor, and the equation becomes

$$\partial_t f + V(e^s)\partial_\theta f = \nu e^{-2s}(\partial_\theta^2 + \partial_s^2) f, \qquad (\theta, s) \in \mathbb{T} \times \mathbb{R}.$$

If we localize near a radius $r_0 = e^{s_0}$ where $V'(r_0) \neq 0$, then to leading order, and up to constants, we recover the equation for the Couette flow. In this sense, the Couette flow can be seen as a first local approximation of a radial vortex in $\mathbb{R}^2$, in a region where the angular velocity is strictly monotone. Below is a video showing the effect of the conformal mapping.



Video. Left: a scalar transported and diffused by the circular flow with $V(r) = 1/(1 + r^2)$. Right: the same solution in log-polar coordinates, where the flow is a shear flow.
The video plays in the online edition of this page. How it was made: page Numerical methods, record `num-log-polar`.

1 The result

Before introducing the main theorem, it is useful to observe that, taking the x-average, one is left with the standard heat equation on $\mathbb{R}$. Namely,

$$\langle f \rangle_x = \frac{1}{2\pi} \int_{\mathbb{T}} f(x, y)\, dx \quad \Longrightarrow \quad \langle f(t) \rangle_x = e^{\nu t \partial_{yy}} \langle f^{\mathrm{in}} \rangle_x.$$

For this part of the solution there is therefore nothing new to say, and it enjoys the standard diffusive estimates. Hence, the quantity to measure is how fast the solution approaches its x-average.

Theorem 1. *Let $\nu \in [0, 1)$ and let f be the solution with initial datum $f^{\mathrm{in}} \in L^2(\mathbb{T} \times \mathbb{R})$. There exist constants $c, C > 0$, independent of ν, such that for all $t \geq 0$ the following inequality holds true:*

$$\|(f - \langle f \rangle_x)(t)\|_{L^2} \leq C e^{-c\nu^{1/3} t} \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{L^2}.$$

Moreover, if $f^{\mathrm{in}} \in H^1(\mathbb{T} \times \mathbb{R})$, the following mixing estimate, uniform in ν, holds true:

$$\|(f - \langle f \rangle_x)(t)\|_{H^{-1}} \leq \frac{C}{1 + t} e^{-c\nu^{1/3} t} \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{H^1}.$$

The x-average as a projection. The x-average is in fact nothing more than the orthogonal projection in L^2 onto the kernel of the transport operator $y\partial_x$: the functions in the kernel

are exactly those that do not depend on x. When the transport operator has a non-trivial kernel, we cannot expect the elements of the kernel to experience any mixing at all from the transport. The interaction with the dissipation is however much more subtle, and it relies heavily on the commutation properties of this projection with the diffusion operator. In this very simple case everything trivializes: the projection commutes with $\nu\Delta$, so the x-average solves the heat equation and $f - \langle f \rangle_x$ solves the same equation as f. But the mathematical structure behind the convergence towards the x-average lies in the fact that the transport operator has a non-trivial kernel.

Why negative Sobolev norms measure mixing. The L^2 norm does not see mixing: for $\nu = 0$ it is conserved, whatever the flow does. The norm of H^{-1} does, because it measures how much of a function has been sent to high frequencies: in Fourier, it weighs each mode by the inverse of the size of its frequency, so it is small when the function lives at high frequencies, even if its L^2 norm is not. On the physical side, negative Sobolev norms scale as powers of a length: the ratio $\|g\|_{H^{-1}}/\|g\|_{L^2}$ can be thought of as the length at which g oscillates. For $\nu = 0$, the theorem says that this length decays like $1/t$. See also the note on mixing in the Overview.

Other function spaces. The theorem is stated in spaces based on L^2. Since the flow is incompressible, the equation does not increase any L^p norm. For $1 < p < \infty$, enhanced dissipation in L^p then follows from the L^2 estimate by interpolation, with a rate that degenerates as $p \to 1$ and as $p \to \infty$. Some techniques give the estimate in L^p directly, without this loss. For the Couette flow, the explicit solution gives the first estimate of the theorem with L^p in place of L^2 for every $p \in [1, \infty]$, with constants independent of p.

Idea. The key idea underlying all the methods below is that the transport operator $y\partial_x$ does not commute with the diffusion $\nu\Delta$: indeed, $[\Delta, y\partial_x] = 2\partial_x\partial_y$. If the two commuted, the solution would be $e^{-ty\partial_x}e^{\nu t\Delta}f^{\text{in}}$ and, since the transport preserves the L^2 norm, the flow would have no effect on the decay. We will highlight in each method where this lack of commutation enters.

At first sight, operators that do not commute look like an annoying technical problem. This is a typical instance where a seemingly unfavorable property, looked at in the right way, turns out to be exactly what helps.

2 Notation and setting

Throughout, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$. For the Fourier transform in x we write

$$f(t,x,y) = \sum_{k\in\mathbb{Z}} f_k(t,y)e^{ikx}, \qquad f_k(t,y) = \frac{1}{2\pi}\int_{\mathbb{T}} f(t,x,y)e^{-ikx}\,dx,$$

so that $f_0 = \langle f \rangle_x$ and $\|f\|_{L^2}^2 = 2\pi\sum_k \|f_k\|_{L^2(\mathbb{R})}^2$. Note that in this model each k-mode evolves independently of the others:

$$\partial_t f_k + iky f_k = \nu(\partial_{yy} - k^2)f_k, \qquad k \in \mathbb{Z}.$$

The Sobolev norms in the theorem are $\|g\|_{H^1}^2 = \|g\|_{L^2}^2 + \|\nabla g\|_{L^2}^2$ and its dual,

$$\|g\|_{H^{-1}} = \sup\Big\{ \int_{\mathbb{T}\times\mathbb{R}} g\varphi\,dx\,dy \;:\; \|\varphi\|_{H^1} \le 1\Big\}.$$

When there is no ambiguity, $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$ denote the norm and the scalar product of $L^2(\mathbb{T}\times\mathbb{R})$ or, for a single mode, of $L^2(\mathbb{R})$.

3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results.*

The estimates mode by mode. Since each k-mode evolves independently, the estimates of the theorem can be stated for each mode separately, keeping track of the dependence on k. There exist constants $c, C > 0$, independent of ν and k, such that for all $k \neq 0$ and $t \geq 0$

$$\|f_k(t)\|_{L^2(\mathbb{R})} \leq Ce^{-c\nu^{1/3}|k|^{2/3}t}\|f_k^{\text{in}}\|_{L^2(\mathbb{R})},$$

$$\|f_k(t)\|_{H^{-1}(\mathbb{R})} \leq \frac{C}{1+|k|t}e^{-c\nu^{1/3}|k|^{2/3}t}\|f_k^{\text{in}}\|_{H^1(\mathbb{R})},$$

where $H^{\pm 1}(\mathbb{R})$ are the Sobolev norms in y only. Higher modes are therefore mixed and dissipated faster. The theorem follows by summing over $k \neq 0$, since $|k| \geq 1$.

3.1 Explicit solution

In this proof we take full advantage of the simplicity of the mathematical setting, which is also what allowed Kelvin and Kolmogorov to write down the solution explicitly.

Proof by the explicit solution. For the equation at mode k, we can also take the Fourier transform in y, with the notation

$$\hat{f}_k(t,\eta) = \int_{\mathbb{R}} f_k(t,y)e^{-i\eta y}\,dy, \qquad f_k(t,y) = \frac{1}{2\pi}\int_{\mathbb{R}} \hat{f}_k(t,\eta)e^{i\eta y}\,d\eta,$$

so that $\|f\|_{L^2}^2 = \sum_{k\in\mathbb{Z}} \|\hat{f}_k\|_{L^2(\mathbb{R})}^2$. By elementary properties of the Fourier transform (multiplication by y becomes $i\partial_\eta$), the equation becomes

$$\partial_t \hat{f}_k - k\partial_\eta \hat{f}_k = -\nu(k^2+\eta^2)\hat{f}_k.$$

This is an advection equation in Fourier, with a damping term. We can then follow the characteristics in η, which are the lines $\eta + kt = \text{const}$, and deduce that the explicit solution is given by

$$\hat{f}_k(t,\eta) = \exp\Big(-\nu\int_0^t (k^2 + (\eta+k\tau)^2)\,d\tau\Big)\,\hat{f}_k^{\text{in}}(\eta+kt).$$

Where the non-commutation enters. In Fourier, the transport is the translation $\eta \mapsto \eta + kt$ and the diffusion is the multiplication by $-\nu(k^2+\eta^2)$. The lack of commutation is simply reflected in the fact that the damping is not constant along the characteristics:

$$\frac{d}{d\tau}\Big[-(k^2+(\eta+k\tau)^2)\Big] = -2k(\eta+k\tau),$$

which is exactly the symbol of the commutator $[\Delta, y\partial_x] = 2\partial_x\partial_y$ at the frequency $(k, \eta+k\tau)$. If the two operators commuted, the damping would be constant along the characteristics, the integral in the exponent would be trivial, and we would only recover the decay of the heat equation, as in the idea above. Instead, the frequency in y grows linearly in time along the characteristics, so the damping grows like $\nu k^2\tau^2$ and its integral like $\nu k^2 t^3$.

The only thing left is therefore to estimate the integral in the exponent, for which we claim that

$$\int_0^t (k^2 + (\eta+k\tau)^2)\,d\tau \geq k^2 t + \frac{1}{12}k^2t^3 \qquad \text{for all } \eta \in \mathbb{R}.$$

Proof of the claim. Computing the integral and completing the square in η,

$$\int_0^t (\eta + k\tau)^2 \, d\tau = \eta^2 t + \eta k t^2 + \frac{1}{3} k^2 t^3 = t\Big(\eta + \frac{kt}{2}\Big)^2 + \frac{1}{12} k^2 t^3 \geq \frac{1}{12} k^2 t^3,$$

with equality only for $\eta = -kt/2$. □

Therefore, using that $f - \langle f \rangle_x = \sum_{k \in \mathbb{Z} \setminus \{0\}} f_k e^{ikx}$, by Plancherel, the fact that $|k| \geq 1$ and that $\nu t^3 \geq 3\nu^{1/3} t - 2$, we get

$$\begin{aligned} \|(f - \langle f \rangle_x)(t)\|_{L^2}^2 &= \sum_{k \neq 0} \int_{\mathbb{R}} e^{-2\nu \int_0^t (k^2 + (\eta + k\tau)^2) \, d\tau} \, |\hat{f}_k^{\mathrm{in}}(\eta + kt)|^2 \, d\eta \\ &\leq C^2 e^{-2c\nu^{1/3} t} \, \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{L^2}^2. \end{aligned}$$

This proves the desired enhanced dissipation estimate. The constants can be explicitly computed.

The constants. Indeed, the claim and $|k| \geq 1$ give the more precise bound

$$\|(f - \langle f \rangle_x)(t)\|_{L^2}^2 \leq e^{-2\nu t - \frac{1}{6}\nu t^3} \, \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{L^2}^2.$$

Moreover, $s^3 - 3s + 2 = (s-1)^2(s+2) \geq 0$ for $s \geq 0$, which for $s = \nu^{1/3} t$ is the inequality $\nu t^3 \geq 3\nu^{1/3} t - 2$. Hence $e^{-\frac{1}{6}\nu t^3} \leq e^{\frac{1}{3}} e^{-\frac{1}{2}\nu^{1/3} t}$, and one can take $c = 1/4$ and $C = e^{1/6}$.

For the uniform mixing estimate, one can directly use the characterization of the Sobolev norms in Fourier, namely

$$\|g\|_{H^s}^2 = \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}} (1 + k^2 + \eta^2)^s \, |\hat{g}_k(\eta)|^2 \, d\eta,$$

which for $s = \pm 1$ are the norms of the theorem. Since $|k| \geq 1$ and $|k| t \leq |\eta| + |\eta + kt|$, by Cauchy–Schwarz we have

$$\begin{aligned} (1+t)^2 \leq (1 + |k| t)^2 &\leq \big(1 + |\eta| + |\eta + kt|\big)^2 \leq (2 + \eta^2)\big(2 + (\eta + kt)^2\big) \\ &\leq (1 + k^2 + \eta^2)\big(1 + k^2 + (\eta + kt)^2\big). \end{aligned}$$

Therefore, bounding the exponential as before,

$$\begin{aligned} \|(f - \langle f \rangle_x)(t)\|_{H^{-1}}^2 &= \sum_{k \neq 0} \int_{\mathbb{R}} \frac{e^{-2\nu \int_0^t (k^2 + (\eta + k\tau)^2) \, d\tau}}{1 + k^2 + \eta^2} \, |\hat{f}_k^{\mathrm{in}}(\eta + kt)|^2 \, d\eta \\ &\leq \frac{C^2 e^{-2c\nu^{1/3} t}}{(1+t)^2} \sum_{k \neq 0} \int_{\mathbb{R}} \big(1 + k^2 + (\eta + kt)^2\big) |\hat{f}_k^{\mathrm{in}}(\eta + kt)|^2 \, d\eta \\ &= \frac{C^2 e^{-2c\nu^{1/3} t}}{(1+t)^2} \, \|f^{\mathrm{in}} - \langle f^{\mathrm{in}} \rangle_x\|_{H^1}^2, \end{aligned}$$

which is the uniform mixing estimate.

In fact, the mode-by-mode estimates can be obtained in the same way, by keeping track of k.

The solution in physical space. For $\nu > 0$, inverting the Fourier transform in y in the

explicit solution is a Gaussian integral, which gives

$$f_k(t,y) = e^{-\nu k^2 t - \frac{1}{12}\nu k^2 t^3} \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\, e^{-\frac{(y-y')^2}{4\nu t}}\, e^{-ikt\frac{y+y'}{2}}\, f_k^{\mathrm{in}}(y')\, dy'.$$

Summing over k, by the Poisson summation formula we get

$$f(t,x,y) = \int_{\mathbb{T}\times\mathbb{R}} G_t(x,y,x',y') f^{\mathrm{in}}(x',y')\, dx'dy',$$

with

$$G_t(x,y,x',y') = \frac{1}{\sqrt{4\pi\nu t}}\, e^{-\frac{(y-y')^2}{4\nu t}} \sum_{n\in\mathbb{Z}} \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\Big(-\frac{\big(x-x'-t\frac{y+y'}{2}+2\pi n\big)^2}{2\sigma_t^2}\Big),$$
$$\sigma_t^2 = 2\nu t + \frac{1}{6}\nu t^3.$$

Without the diffusion in x, namely with $\sigma_t^2 = \frac{1}{6}\nu t^3$, and on the whole plane, namely keeping only the term $n = 0$, this is the fundamental solution written by Kolmogorov (1934). In the variable x the solution spreads over the length $\sqrt{\nu t^3}$, much larger than the diffusive one $\sqrt{\nu t}$. For $t \gg \nu^{-1/3}$ this length exceeds the size of the torus and the kernel is almost independent of x: this is the enhanced dissipation seen in physical space.

□

3.2 Hypocoercivity

This proof builds on elementary techniques, in the spirit of the hypocoercivity theory of Villani (2009): it only uses energy estimates, namely integrations by parts and the Cauchy–Schwarz inequality. The key point is that the transport does not commute with the operator that generates the dissipation, which here is, morally, ∂_y.

Proof by hypocoercivity. We first simplify the setting by ignoring the diffusion in x. In fact, since ∂_{xx} commutes with all the other operators in the equation, we have $f = e^{\nu t\partial_{xx}} g$, where g solves the equation without the diffusion in x, with $g^{\mathrm{in}} = f^{\mathrm{in}}$, and $e^{\nu t\partial_{xx}}$ does not increase the L^2 norm. It is therefore enough, and somewhat more instructive, to prove the estimate for

$$\partial_t g + y\partial_x g = \nu\partial_{yy} g,$$

with $\nu > 0$, since for $\nu = 0$ there is nothing to prove. Without loss of generality, we also assume that g has zero x-average, since the x-average solves a decoupled equation and a zero x-average is preserved in time. Note that on functions with zero x-average we have the Poincaré inequality in x, namely $\|h\| \le \|\partial_x h\|$.

We now look at the equations satisfied by the derivatives of g. The derivative in x commutes with everything, so $\partial_x g$ solves the same equation as g. The derivative in y does not commute with the transport, since $[\partial_y, y\partial_x] = \partial_x$, and we get

$$\partial_t(\partial_x g) + y\partial_x(\partial_x g) = \nu\partial_{yy}(\partial_x g),$$
$$\partial_t(\partial_y g) + y\partial_x(\partial_y g) = \nu\partial_{yy}(\partial_y g) - \partial_x g.$$

This is a coupled system for $\partial_x g$ and $\partial_y g$, and there is no direct dissipation for $\partial_x g$. However, the last term acts as a forcing in the equation for $\partial_y g$, and this is exactly the effect of the non-commutation between the transport and ∂_y.

Relation to hyperbolic–parabolic systems. Systems of this type, in which only part of the unknowns is directly dissipated, are the typical situation, for instance, in compressible fluids or for the Boltzmann equation. A general framework for such hyperbolic–parabolic systems was developed by Kawashima (1983) in his PhD thesis (see also the references therein). In fact, hypocoercivity and many other modern energy methods have analogies with the techniques employed by Kawashima.

Since the transport is antisymmetric, the energy identity for the first equation is

$$\frac{1}{2}\frac{d}{dt}\|\partial_x g\|^2 = -\nu\|\partial_y\partial_x g\|^2.$$

By the Poincaré inequality in x, the right-hand side clearly controls $\|\partial_y g\|^2$, but it does not directly control $\|\partial_x g\|^2$, because there is no Poincaré inequality when $y \in \mathbb{R}$. To get back some dissipation, one of the key ideas behind hypocoercivity is to exploit mixed inner products. Here,

$$\frac{d}{dt}\langle \partial_x g, \partial_y g\rangle = -\|\partial_x g\|^2 + \text{terms coming from the dissipation.}$$

The game is then to tailor an energy functional containing all the quantities needed to conclude:

- $\|\partial_x g\|$, which is what we want to decay;
- $\langle \partial_x g, \partial_y g\rangle$, which gives the dissipation of $\partial_x g$;
- $\|\partial_y g\|$, which is needed to control the mixed product.

For $\alpha, \beta > 0$ to be chosen, we define

$$E(t) = \frac{1}{2}\Big(\|\partial_x g\|^2 + \alpha\|\partial_y g\|^2 + 2\beta\langle \partial_x g, \partial_y g\rangle\Big).$$

The identities for the last two terms are

$$\frac{1}{2}\frac{d}{dt}\|\partial_y g\|^2 = -\nu\|\partial_{yy} g\|^2 - \langle \partial_x g, \partial_y g\rangle, \qquad \frac{d}{dt}\langle \partial_x g, \partial_y g\rangle = -\|\partial_x g\|^2 - 2\nu\langle \partial_x\partial_y g, \partial_{yy} g\rangle,$$

and therefore

$$\frac{d}{dt}E = -\nu\|\partial_x\partial_y g\|^2 - \alpha\nu\|\partial_{yy} g\|^2 - \beta\|\partial_x g\|^2 - \alpha\langle \partial_x g, \partial_y g\rangle - 2\beta\nu\langle \partial_x\partial_y g, \partial_{yy} g\rangle.$$

The first three terms have a good sign, and the third one is the dissipation of $\partial_x g$ that was missing. The last two have no sign, and must be absorbed by the good ones. By the Cauchy–Schwarz and Young inequalities, and by the Poincaré inequality in x for the second bound,

$$2\beta\nu|\langle \partial_x\partial_y g, \partial_{yy} g\rangle| \le \frac{\nu}{2}\|\partial_x\partial_y g\|^2 + 2\beta^2\nu\|\partial_{yy} g\|^2,$$
$$\alpha|\langle \partial_x g, \partial_y g\rangle| \le \frac{\beta}{2}\|\partial_x g\|^2 + \frac{\alpha^2}{2\beta}\|\partial_x\partial_y g\|^2.$$

Therefore, if

$$2\beta^2 \le \alpha \qquad \text{and} \qquad 2\alpha^2 \le \beta\nu,$$

we obtain

$$\frac{d}{dt}E \le -\frac{\beta}{2}\|\partial_x g\|^2 - \frac{\nu}{4}\|\partial_x\partial_y g\|^2.$$

Moreover, the first condition guarantees that E is comparable to the sum of its first two terms:

$$\frac{1}{4}\|\partial_x g\|^2 \le E \le \frac{3}{4}\|\partial_x g\|^2 + \alpha\|\partial_y g\|^2.$$

The two conditions together force $8\beta^3 \le \nu$. Thus, the dissipation that we can obtain for $\partial_x g$ is at most of order $\nu^{1/3}$, and this is where the rate comes from. We then choose

$$\beta = \frac{1}{2}\nu^{1/3}, \qquad \alpha = \frac{1}{2}\nu^{2/3}.$$

Using once more the Poincaré inequality in x,

$$\frac{d}{dt}E \le -\frac{\nu^{1/3}}{4}\Big(\|\partial_x g\|^2 + \nu^{2/3}\|\partial_y g\|^2\Big) \le -\frac{\nu^{1/3}}{3}E,$$

and hence, for all $t \ge s \ge 0$,

$$\|\partial_x g(t)\|^2 \le 4E(t) \le 4e^{-\frac{1}{3}\nu^{1/3}(t-s)}E(s).$$

The choice of the coefficients. The powers of ν can be guessed as follows. The transport turns derivatives in x into derivatives in y. For $\nu = 0$ the solution is $g^{\mathrm{in}}(x - ty, y)$, so that $\partial_y g \approx -t\partial_x g$ for large times. The dissipation is then expected to satisfy $\nu\|\partial_y g\|^2 \approx \nu t^2\|\partial_x g\|^2$. Once integrated up to a time τ, this is comparable to $\|\partial_x g\|^2$ when $\nu\tau^3 \approx 1$, that is $\tau = \nu^{-1/3}$. At that time the three terms of E have sizes 1, $\alpha\tau^2$ and $\beta\tau$ relative to $\|\partial_x g\|^2$, and they are comparable exactly when $\alpha \approx \nu^{2/3}$ and $\beta \approx \nu^{1/3}$. In other words, E measures ∂_y in units of $\nu^{1/3}$.

It remains to remove $\|\partial_y g^{\mathrm{in}}\|$, which appears in $E(0)$ and is not controlled by the norms stated in the theorem. For this we use the smoothing effect of the diffusion. Integrating in time the energy identity for $\partial_x g$, and using the Poincaré inequality in x, we get

$$\nu\int_0^T \|\partial_y g(s)\|^2\,ds \le \frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2 \qquad \text{for all } T > 0.$$

We then take $T = \nu^{-1/3}$, the time scale of the decay. Hence there is a time $t_0 \in (0, \nu^{-1/3}]$ such that $\nu^{2/3}\|\partial_y g(t_0)\|^2 \le \frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2$ and therefore, since $\|\partial_x g\|$ does not increase,

$$E(t_0) \le \frac{3}{4}\|\partial_x g(t_0)\|^2 + \frac{1}{2}\nu^{2/3}\|\partial_y g(t_0)\|^2 \le \|\partial_x g^{\mathrm{in}}\|^2.$$

For $t \ge t_0$ we use the decay of E starting from the time t_0, while for $t \le t_0$ we simply use that $\|\partial_x g\|$ does not increase. Since $\nu^{1/3}t_0 \le 1$, in both cases

$$\|\partial_x g(t)\| \le 2e^{1/6}e^{-\frac{1}{6}\nu^{1/3}t}\|\partial_x g^{\mathrm{in}}\| \qquad \text{for all } t \ge 0.$$

Finally, since ∂_x commutes with the equation and is invertible on functions with zero x-average, the same estimate holds with g in place of $\partial_x g$, and hence for f, since $\|f(t)\| \le \|g(t)\|$ and $g^{\mathrm{in}} = f^{\mathrm{in}}$. This proves the enhanced dissipation estimate of the theorem.

Avoiding the smoothing step. The smoothing step can be avoided by letting the coefficients depend on time, exactly as one proves the gain of regularity for the heat equation through energy estimates with the weight t in front of $\|\partial_y g\|^2$. With

$$q(t) = \frac{\nu^{1/3}t}{1 + \nu^{1/3}t}, \qquad \alpha(t) = a\,\nu^{2/3}q(t), \qquad \beta(t) = b\,\nu^{1/3}q(t)^2,$$

the functional E at the initial time is just $\frac{1}{2}\|\partial_x g^{\mathrm{in}}\|^2$, so nothing is asked of $\partial_y g^{\mathrm{in}}$. For $t \ll \nu^{-1/3}$ the weights behave like νt and νt^2, as for the heat equation, and for $t \gg \nu^{-1/3}$ they are the constant ones used above. The price is two new terms, coming from the time

derivatives of the weights, which are absorbed by the good terms we already have if a and b are small enough (we leave this as an exercise). A functional of this kind measures the solution at each time at the right scale, and gives the decay and the gain of regularity at once.

Uniform mixing estimate. It remains to prove the uniform mixing estimate. We go back to f, which has zero x-average, and use a vector field that commutes with the equation, namely

$$Z := \partial_y + t\partial_x.$$

Indeed, $[\partial_t + y\partial_x, Z] = \partial_x - \partial_x = 0$, and Z has constant coefficients in x and y, so it also commutes with $\nu\Delta$. For $\nu = 0$ we have $Zf(t,x,y) = (\partial_y f^{\mathrm{in}})(x - ty, y)$: Z is the derivative in y in the coordinates that move with the flow. Hence Zf solves exactly the same equation as f, with datum $\partial_y f^{\mathrm{in}}$ and zero x-average, and the enhanced dissipation estimate gives

$$\|Zf(t)\| \le 2e^{1/6} e^{-\frac{1}{6}\nu^{1/3}t} \|\partial_y f^{\mathrm{in}}\|.$$

To bound the H^{-1} norm we argue by duality. Let $\varphi \in H^1(\mathbb{T}\times\mathbb{R})$. Since f has zero x-average, we can replace φ by $\varphi - \langle\varphi\rangle_x = \partial_x\Phi$, where Φ is the primitive in x with zero x-average. By the Poincaré inequality in x, $\|\Phi\| \le \|\varphi\|$ and $\|\partial_y\Phi\| \le \|\partial_y\varphi\|$. Integrating by parts and using $t\partial_x = Z - \partial_y$,

$$t\int_{\mathbb{T}\times\mathbb{R}} f\varphi\,dx\,dy = -\int_{\mathbb{T}\times\mathbb{R}} (t\partial_x f)\Phi\,dx\,dy = -\int_{\mathbb{T}\times\mathbb{R}} (Zf)\Phi\,dx\,dy - \int_{\mathbb{T}\times\mathbb{R}} f\,\partial_y\Phi\,dx\,dy.$$

Therefore $t\|f(t)\|_{H^{-1}} \le \|Zf(t)\| + \|f(t)\|$. Since also $\|f(t)\|_{H^{-1}} \le \|f(t)\|$, the enhanced dissipation estimates for f and Zf give

$$(1+t)\|f(t)\|_{H^{-1}} \le 2\|f(t)\| + \|Zf(t)\| \le 6e^{1/6} e^{-\frac{1}{6}\nu^{1/3}t} \|f^{\mathrm{in}}\|_{H^1},$$

which is the uniform mixing estimate.

The mode-by-mode estimates can be obtained in the same way: on a single mode k the Poincaré inequality in x becomes the identity $\|\partial_x h\| = |k|\|h\|$, and the argument can be repeated with νk^2 in place of ν in the choice of α, β and T, which gives the rate $\nu^{1/3}|k|^{2/3}$. □

General theory: Hypocoercivity.

3.3 Resolvent method

This proof builds on resolvent estimates for the operator $-\nu\partial_{yy} + iky$, uniform along the imaginary axis. In fact, estimates on the decay of the associated semigroup are tied to resolvent bounds by Gearhart–Prüss type theorems, in the spirit of Helffer and Sjöstrand (2010) and Wei (2021). The key point is that ∂_y does not commute with the multiplication by y, so a function cannot concentrate near the critical layer, where iky equals the spectral parameter, unless of course its derivative is large, in which case the dissipation kicks in.

Proof by the resolvent method. We fix $k \neq 0$ and write $f_k = e^{-\nu k^2 t} g$. Then the function g solves

$$\partial_t g + H_k g = 0, \qquad H_k := -\nu\partial_{yy} + iky \quad \text{on } L^2(\mathbb{R}).$$

We note that the spectrum of H_k on $L^2(\mathbb{R})$ is in fact empty (it is a complex Airy operator), and we aim instead at estimating the resolvent of H_k. To this end, we define the *pseudospectral abscissa* of H_k as

$$\Psi(H_k) := \inf\{\|(H_k - ik\lambda)g\| \ : \ \lambda\in\mathbb{R},\ g\in D(H_k),\ \|g\| = 1\}.$$

What the pseudospectral abscissa measures. For $\varepsilon > 0$, the ε-pseudospectrum of H_k is the set of $z \in \mathbb{C}$ where the resolvent is large, $\|(H_k - z)^{-1}\| > 1/\varepsilon$. Equivalently, it is the set of z with an approximate eigenfunction, $\|(H_k - z)g\| < \varepsilon\|g\|$. For a self-adjoint operator this is the ε-neighbourhood of the spectrum. For an operator that is not normal, as H_k, it can be much larger, and it is the resolvent, not the spectrum, that controls the semigroup. The pseudospectral abscissa is the largest ε for which the ε-pseudospectrum does not touch the imaginary axis, namely $\Psi(H_k) = \big(\sup_\lambda \|(H_k - ik\lambda)^{-1}\|\big)^{-1}$. For H_k a translation in y only shifts $\operatorname{Im} z$, so the pseudospectra are half-planes. For more on pseudospectra, see Trefethen and Embree (2005).

This is a fundamental quantity, which controls the decay in time of the semigroups generated by accretive operators (H is accretive if $-H$ is dissipative). Indeed, Gearhart–Prüss type bounds link the pseudospectral abscissa to the decay of the semigroup, see Helffer and Sjöstrand (2010), and here we appeal to the result of Wei (2021) (see the Toolkit), which tells us that

$$\|e^{-tH_k}\|_{L^2\to L^2} \le e^{\pi/2}\, e^{-\Psi(H_k)t} \qquad \text{for all } t \ge 0.$$

Therefore, the enhanced dissipation estimate follows from the claim below.

Claim. *There exists $c_0 > 0$, independent of ν and k, such that $\Psi(H_k) \ge c_0\, \nu^{1/3}|k|^{2/3}$.*

Once the claim is proved, the enhanced dissipation estimate of the theorem follows by going back from g to f_k and summing over $|k| \ge 1$.

Proof of the claim. Fix $\lambda \in \mathbb{R}$ and $g \in D(H_k)$, and let

$$F_\lambda := (H_k - ik\lambda)g = -\nu\partial_{yy}g + ik(y-\lambda)g.$$

The goal is to bound $\|g\|$ by $\|F_\lambda\|$, uniformly in λ.

Step 1 (energy identity). The first observation is the standard energy identity, which in this spectral picture reads

$$\operatorname{Re}\big(\langle F_\lambda, g\rangle\big) = \nu\|\partial_y g\|^2, \qquad \text{hence} \qquad \nu\|\partial_y g\|^2 \le \|F_\lambda\|\|g\|.$$

Note that here we have not used the transport operator at all.

Step 2 (using the transport). To use the transport, we test against $(y-\lambda)g$ and take the imaginary part. Integrating by parts once,

$$\operatorname{Im}\big(\langle F_\lambda, (y-\lambda)g\rangle\big) = k\|(y-\lambda)g\|^2 + \nu \operatorname{Im}\big(\langle \partial_y g, g\rangle\big),$$

and therefore

$$|k|\,\|(y-\lambda)g\|^2 \le \|F_\lambda\|\,\|(y-\lambda)g\| + \nu\|\partial_y g\|\|g\|.$$

This controls g away from the critical layer $y = \lambda$, where the transport vanishes. To recover the full norm of g we use a functional inequality, which only requires an integration by parts:

$$\|g\|^2 = \int_{\mathbb{R}} \partial_y(y-\lambda)\,|g|^2\,dy = -2\operatorname{Re}\big(\langle (y-\lambda)g, \partial_y g\rangle\big) \le 2\,\|(y-\lambda)g\|\,\|\partial_y g\|.$$

In words, g cannot concentrate near the critical layer unless its derivative is large.

Step 3 (conclusion). By Step 2 and Young's inequality,

$$\|(y-\lambda)g\|^2 \le \frac{1}{k^2}\|F_\lambda\|^2 + \frac{2\nu}{|k|}\|\partial_y g\|\|g\|.$$

Inserting this in the functional inequality, and using Step 1 to bound $\|\partial_y g\|$,

$$\|g\|^4 \le 4\,\|(y-\lambda)g\|^2\|\partial_y g\|^2 \lesssim \frac{1}{\nu k^2}\|F_\lambda\|^3\|g\| + \frac{1}{\nu^{1/2}|k|}\|F_\lambda\|^{3/2}\|g\|^{5/2}.$$

By Young's inequality once more, $\|g\|^3 \lesssim \frac{1}{\nu k^2}\|F_\lambda\|^3$, that is $\|F_\lambda\| \gtrsim \nu^{1/3}|k|^{2/3}\|g\|$ uniformly in λ.

A more flexible argument: splitting near and away from the critical layer. The functional inequality of Step 2 uses that $y - \lambda$ has derivative one everywhere. For a general shear $v(y)$ this is not available: for instance $\int v'|g|^2$ has no sign if v is not monotone. One then splits the norm of g, for $\delta > 0$ to be chosen.
Near the critical layer the transport is perturbative and only the dissipation is available, but the region is thin:

$$\int_{|y-\lambda|<\delta} |g|^2 \, dy \leq 2\delta \|g\|_{L^\infty}^2 \leq 2\delta \, \|g\| \, \|\partial_y g\|.$$

Away from the critical layer the transport can be inverted:

$$\int_{|y-\lambda|\geq\delta} |g|^2 \, dy \leq \frac{1}{\delta^2} \|(y-\lambda)g\|^2 \leq \frac{1}{k^2\delta^2} \|F_\lambda\|^2 + \frac{2\nu}{|k|\delta^2} \|\partial_y g\| \|g\|.$$

Adding the two,

$$\|g\|^2 \leq \frac{1}{k^2\delta^2} \|F_\lambda\|^2 + 2\Big(\delta + \frac{\nu}{|k|\delta^2}\Big) \|\partial_y g\| \|g\|.$$

The two terms in the bracket balance for $\delta = (\nu/|k|)^{1/3}$, the width of the critical layer, and one concludes with Step 1 as before. This is the structure of the proofs for general shear flows, where the test function away from the layer is usually a regularisation of $\operatorname{sgn}(v - \lambda)$; see Gallay (2022) and Coti Zelati and Gallay (2023).

□

Uniform mixing estimate. It remains to prove the uniform mixing estimate. Here it is convenient to use something that commutes, namely the vector field

$$Z_k := \partial_y + ikt.$$

This is a commuting vector field for the transport equation (namely $[Z_k, \partial_t + iky] = 0$), but in this particular case it also commutes with the dissipation. Hence $Z_k f_k$ solves exactly the same equation as f_k, with datum $\partial_y f_k^{\mathrm{in}}$, and by the enhanced dissipation estimate

$$\|(\partial_y + ikt) f_k(t)\|_{L^2} \leq C e^{-c\nu^{1/3}|k|^{2/3}t} \|\partial_y f_k^{\mathrm{in}}\|_{L^2}.$$

To bound the H^{-1} norm, we take a test function $\varphi \in H^1(\mathbb{R})$ and note that, integrating by parts,

$$ikt \int_{\mathbb{R}} f_k \varphi \, dy = \int_{\mathbb{R}} (Z_k f_k) \varphi \, dy + \int_{\mathbb{R}} f_k \, \partial_y \varphi \, dy.$$

Therefore, since for $|k|t \leq 1$ there is nothing to prove, by the dual definition of the H^{-1} norm and the enhanced dissipation estimates for $Z_k f_k$ and f_k we conclude that, for $|k|t \geq 1$,

$$\|f_k(t)\|_{H^{-1}} \leq \frac{1}{|k|t} \Big(\|Z_k f_k(t)\|_{L^2} + \|f_k(t)\|_{L^2} \Big) \lesssim \frac{1}{1+|k|t} \, e^{-c\nu^{1/3}|k|^{2/3}t} \, \|f_k^{\mathrm{in}}\|_{H^1}.$$

□

General theory: Spectral methods.

3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

General theory: Stochastic methods.

3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

General theory: Hörmander theory.

4 What this example teaches

[What the example shows, and what the general theory keeps of it.]

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